

Principles and Specifications of Concurrent Systems

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The *Principles* and *Specification* Tracks

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1 Introduction

- 1.1 Concurrent Computation
- 1.2 Modeling Computation
- 1.3 Specification
- 1.4 Systems and Languages

Sections colored like this have
not yet been written.

2 The One-Bit Clock

- 2.1 The Clock's Behaviors
- 2.2 Describing the Behaviors
- 2.3 Writing the Specification
- 2.4 The Pretty-Printed Version of Your Spec
- 2.5 Checking the Specification
- 2.6 Computing the Behaviors from the Specification
- 2.7 Other Ways of Writing the Behavior Specification
- 2.8 Specifying the Clock in PlusCal

3 The Die Hard Problem

- 3.1 Representing the Problem in TLA⁺
- 3.2 Applying TLC
- 3.3 Expressing the Problem in PlusCal

4 Euclid's Algorithm

- 4.1 The Greatest Common Divisor
 - 4.1.1 Divisors
 - 4.1.2 CHOOSE and the Maximum of a Set
 - 4.1.3 The GCD Operator
- 4.2 Comments
- 4.3 The Algorithm
- 4.4 The TLA⁺ Translation
- 4.5 Checking Safety
- 4.6 Checking Liveness
- 4.7 The Translation Revisited
- 4.8 The Grain of Atomicity
- 4.9 Why Euclid's Algorithm Is Correct
 - 4.9.1 Proving Invariance
 - 4.9.2 Verifying *GCD1*–*GCD3*

5 The Generalized Die Hard Problem

- 5.1 The PlusCal Representation
- 5.2 Checking the Algorithm
- 5.3 The TLA⁺ Translation

6 Alternation

- 6.1 The Problem
- 6.2 The One-Bit Clock Revisited
- 6.3 Specifying Alternation: Safety
- 6.4 Specifying Alternation: Liveness
- 6.5 The Two-Phase Handshake Protocol
- 6.6 Refinement
- 6.7 Refinement and Stuttering
 - 6.7.1 Adding Steps
 - 6.7.2 Temporal Logic and Stuttering
 - 6.7.3 A Finer-Grained Algorithm
- 6.8 Temporal Logic and Refinement
- 6.9 Alternation Revisited
- 6.10 Round-Robin Synchronization
 - 6.10.1 The One-Bit Clock Revisited Again
 - 6.10.2 An N -Valued Clock
 - 6.10.3 An Implementation of the N -Valued Clock
 - 6.10.4 Round-Robin Synchronization

The *Principles* Track

7 Mutual Exclusion

- 7.1 The Problem
- 7.2 The One-Bit Protocol
 - 7.2.1 The Protocol
 - 7.2.2 An Assertion Proof
 - 7.2.3 Using TLC to Check an Inductive Invariant
- 7.3 The Two-Process One-Bit Algorithm
 - 7.3.1 The Two-Process Algorithm
 - 7.3.2 Busy Waiting Versus Synchronization Primitives
 - 7.3.3 Requirement (c)
- 7.4 Proving Liveness
- 7.5 An Informal Proof
- 7.6 A More Formal Proof
- 7.7 The N -Process One-Bit Algorithm

- 7.8 The Bakery Algorithm
 - 7.8.1 The Big-Step Algorithm
 - 7.8.2 Choosing the Grain of Atomicity
 - 7.8.3 The Atomic Bakery Algorithm
 - 7.8.4 The Real Bakery Algorithm
- 7.9 Mutual Exclusion in Modern Programs

? ← → C I S 8 The Bounded Channel and Bounded Buffer

- 8.1 The Bounded Channel
 - 8.1.1 The Specification
 - 8.1.2 Safety
 - 8.1.3 Liveness
 - 8.1.4 Implementing The Bounded Channel
- 8.2 The Bounded Buffer
 - 8.2.1 Modular Arithmetic
 - 8.2.2 The Algorithm
- 8.3 The Bounded Buffer Implements the Bounded Channel
 - 8.3.1 The Refinement Mapping
 - 8.3.2 Showing Implementation
 - 8.3.3 Liveness
- 8.4 A Finer-Grained Bounded Buffer
- 8.5 Further Refinement
- 8.6 What is a Process?

The *Specification* Track

9 An Input/Output Specification

- 9.1 The Example
- 9.2 Sorting
- 9.3 Votes
- 9.4 The Borda Ranking
- 9.5 The Condorcet Ranking
- 9.6 Transitive Closure
 - 9.6.1 A Mathematical Definition
 - 9.6.2 A Definition TLC Can Execute Faster
 - 9.6.3 Warshall's Algorithm
- 9.7 The Condorcet Ranking Revisited

The TLA⁺ Proof Track

10 About Proofs and Proving

- 10.1 About Proofs

11 Correctness of Euclid's Algorithm

11.1 Proving Safety

11.2 Proving Properties of the GCD

12 The Proof Language

12.1 What a Theorem Asserts

12.2 The Hierarchical Structure of a Proof

12.2.1 Writing Structured Proofs

12.2.2 Reading Structured Proofs

12.3 The State of a Proof

12.3.1 Steps That Can Have a Proof

12.3.2 Steps That Cannot Have a Proof

12.4 Proof Obligations

12.5 Further Details

12.5.1 Additional Language Features

12.5.2 Importing

12.5.3 Recursively Defined Functions and Operators

12.5.4 The Fine Print

13 The Bounded Buffer Proof

Math

13 Arithmetic and Logic

13.1 Arithmetic

13.2 Mathematical Logic

13.3 Propositional Logic

13.3.1 $\wedge$ and $\vee$

13.3.2 Other Propositional Operators

13.4 Predicate Logic

13.5 The CHOOSE Operator

14 Sets

14.1 An Introduction to Sets

14.2 Simple Set Operators

14.3 Set Constructors

14.4 SUBSET and UNION

14.5 Collections too Big to Be Sets

14.6 Bags

15 Functions

- 15.1 Functions and Their Domains
- 15.2 Writing Functions
- 15.3 Sets of Functions
- 15.4 The EXCEPT Construct
- 15.5 Tuples
- 15.6 Records
- 15.7 Strings

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16 Miscellaneous Constructs

- 16.1 Conditional Constructs
 - 16.1.1 IF / THEN / ELSE
 - 16.1.2 CASE
- 16.2 Definitions
 - 16.2.1 Simple Operator Definitions
 - 16.2.2 Function Definitions
 - 16.2.3 Recursive Operator Definitions
 - 16.2.4 Recursive Or Inductive?
- 16.3 The LET / IN Construct
- 16.4 The LAMBDA Construct

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17 Temporal Logic

- 17.1 Understanding Temporal Formulas
- 17.2 Proof Rules and Proofs
- 17.3 Rules for Proving Safety
- 17.4 Leads To
 - 17.4.1 The Leads-To Induction Rule
 - 17.4.2 The $\square \leadsto$ Rule
 - 17.4.3 Proving $\leadsto$ Formulas by Contradiction
- 17.5 Fairness
 - 17.5.1 The ENABLED Operator
 - 17.5.2 Weak Fairness
 - 17.5.3 Strong Fairness
 - 17.5.4 Proving $\leadsto$ Properties with Fairness
 - 17.5.5 Proving Fairness

Topics

18 Variable Hiding and Auxiliary Variables

19 Reduction

20 Debugging With TLC

20.1 Print Statements

20.2 Having TLC Set and Read Values

20.3 Using LET

20.4 The Perils of Lazy Evaluation

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